

Experiment No.: 6

- Title:** use string functions and regex for email validation, data extraction, and text analysis.
- Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- Theory:** String functions in JavaScript are used to manipulate and analyze text. In this practical, functions like `toUpperCase()`, `toLowerCase()`, `includes()`, `match()`, `split()`, and `length` are used for text analysis.

Regular Expressions (Regex): are patterns used to search, extract, and validate text. In this practical, Regex is used to extract email IDs from a paragraph and check whether they follow a valid email format.

The program also checks the presence of "Welcome", counts words and the occurrence of "email", and displays valid and invalid email IDs.

4. Output :

File C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaaji%20javascript/P6%20email%20validation.html

Email Validation & Text Analysis

Enter a paragraph to extract and validate email IDs

Welcome to our college project portal! Hello, I am Ananya, a college student working on a web development project. I usually communicate with my classmates through email. My friend Rahul uses rahul.sharma@gmail.com, while Priya can be contacted at priya123@yahoo.com. Aman gave me his email as aman@company, which is invalid because it has no proper domain extension. Neha accidentally wrote neha@.com, and Rohan entered rohan@gmail.com, both of which are invalid email formats. I also received valid email IDs such as karan.dev@outlook.com and sneha2026@proton.me. However, vikas@domain..com and meena@gmail.com are invalid. Welcome to our platform, where we share assignments, project files, and important college updates through email.

Analyze Text

 **Extracted Emails**

rahul.sharma@gmail.com, priya123@yahoo.com, aman@company, neha@.com, karan.dev@outlook.com, sneha2026@proton.me, vikas@domain..com

 **Email Validation**

false

 **Valid Emails**

rahul.sharma@gmail.com → Valid

priya123@yahoo.com → Valid

karan.dev@outlook.com → Valid

sneha2026@proton.me → Valid

vikas@domain..com → Valid



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

C:/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/P6%20email%20validation.html

Invalid Emails

aman@company → Invalid

neha@.com → Invalid

Uppercase

WELCOME TO OUR COLLEGE PROJECT PORTAL! HELLO, I AM ANANYA, A COLLEGE STUDENT WORKING ON A WEB DEVELOPMENT PROJECT. I USUALLY COMMUNICATE WITH MY CLASSMATES THROUGH EMAIL. MY FRIEND RAHUL USES RAHUL.SHARMA@GMAIL.COM, WHILE PRIYA CAN BE CONTACTED AT PRIYA123@YAHOO.COM. AMAN GAVE ME HIS EMAIL AS AMAN@COMPANY, WHICH IS INVALID BECAUSE IT HAS NO PROPER DOMAIN EXTENSION. NEHA ACCIDENTALLY WROTE NEHA@.COM, AND ROHAN ENTERED ROHAN GMAIL.COM, BOTH OF WHICH ARE INVALID EMAIL FORMATS. I ALSO RECEIVED VALID EMAIL IDS SUCH AS KARAN.DEV@OUTLOOK.COM AND SNEHA2026@PROTON.ME. HOWEVER, VIKAS@DOMAIN.COM AND MEENA#GMAIL.COM ARE INVALID. WELCOME TO OUR PLATFORM, WHERE WE SHARE ASSIGNMENTS, PROJECT FILES, AND IMPORTANT COLLEGE UPDATES THROUGH EMAIL.

Lowercase

welcome to our college project portal! hello, i am ananya, a college student working on a web development project. i usually communicate with my classmates through email. my friend rahul uses rahul.sharma@gmail.com, while priya can be contacted at priya123@yahoo.com. aman gave me his email as aman@company, which is invalid because it has no proper domain extension. neha accidentally wrote neha@.com, and rohan entered rohan gmail.com, both of which are invalid email formats. i also received valid email ids such as karan.dev@outlook.com and sneha2026@proton.me. however, vikas@domain.com and meena#gmail.com are invalid. welcome to our platform, where we share assignments, project files, and important college updates through email.

Text Analysis

738

Text Length

105

No. of Words

5

"email" Occurrence

Contains "Welcome"

true

5. Result/Conclusion: The program was successfully executed. It extracts email IDs from the given paragraph, validates them using Regex, and displays valid and invalid emails separately. It also performs uppercase/lowercase conversion, checks for "Welcome", calculates text length, counts words, and finds the occurrence of "email". Thus, string functions and regular expressions were successfully used for email validation, data extraction, and text analysis.



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Case Study No.: 06

1. Case Study Title: To perform basic string operations ; to reverse a string and count number of vowels in a paragraph.

2. Case Study Program Code:

<> Pract 6 case study.html > html > body

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>String Operations</title>
5 <style>
6   body {
7     background-color: rgb(171, 238, 243);
8     font-family: Arial;
9   }
10  input {
11    padding: 8px;
12    margin: 5px;
13    width: 350px;
14  }
15  textarea {
16    padding: 10px;
17    margin: 5px;
18    width: 500px;
19    height: 80px;
20  }
21  button {
22    padding: 10px 20px;
23    margin: 10px 5px;
24  }
25  #output {
26    margin-top: 20px;
27    font-size: 18px;
28  }
29 </style>
30 </head>
31 <body>
```

```
30 </head>
31 <body>
32 <h2>Vithika's JavaScript String Operations</h2>
33 <label>Enter String:</label><br>
34 <textarea id="str" placeholder="Enter your string or paragraph">
35 <br>
36 <label>Substring Start:</label>
37 <input type="number" id="start" placeholder="Enter start index">
38 <br>
39 <label>Substring End:</label>
40 <input type="number" id="end" placeholder="Enter end index">
41 <br>
42 <label>IndexOf - Enter text to search:</label>
43 <input type="text" id="search" placeholder="Enter character to search">
44 <br>
45 <label>Split - Enter separator:</label>
46 <input type="text" id="separator" placeholder="Example: space">
47 <br>
48 <label>Replace - Text to replace:</label>
49 <input type="text" id="oldText" placeholder="Enter text to replace">
50 <br>
51 <label>Replace - New text:</label>
52 <input type="text" id="newText" placeholder="Enter new text">
53 <br>
54 <button onclick="performOperations()">Perform Operations</button>
55 <div id="output"></div>
56 <script src="shreya_case6.js"></script>
57 </body>
58 </html>
```

3.Output:

Vithika's JavaScript String Operations

Enter String:

```
Technology is changing the way people live, work, and communicate.  
The internet provides easy access to information from different  
parts of the world.  
Learning new technologies can improve problem-solving and  
creativity.
```

Substring Start:

Substring End:

IndexOf - Enter text to search:

Split - Enter separator:

Replace - Text to replace:

Replace - New text:

Perform Operations

4.Result/Conclusion:

The JavaScript string operations such as substring, indexOf, split, and replace were successfully performed on the given string. The practical helped in understanding how JavaScript string methods can be used to extract, search, split, and replace text efficiently.